MarketBite: Dynamic Pricing for Campus Dining

Executive Summary for The Heist Hackathon

I. Problem and Solution

Prepared food can remain unsold at campus cafes while students seek affordable meals. MarketBite connects this surplus with students through staff-approved price reductions. We propose starting at Connections Cafe at the University of Rochester. A live food-market display and mobile menu would help students find current offers, helping cafes sell eligible food before cutoff while respecting cafe price floors and making meals more affordable.

II. Sustainability Vault

MarketBite enters the Sustainability vault to help cafes sell surplus food before it goes unused. Our "heist" is to recover value from food left unsold by making timely offers visible to students. The goal is to make better use of the ingredients, energy, and staff time invested in each meal. A campus pilot would test whether more meals are sold and less edible food is discarded.

III. Key Features

The prototype has three main displays. The Food Display presents available items and prices. The Exchange Display shows prices and market-style charts. The Mobile Display lets students browse offers and make simulated purchases.

Cafe Desk is the shared control panel for cafe owners. Owners set inventory, hours, price limits, and promotions, then run the same simulation across all three displays. Each view reflects the same confirmed prices and availability. Purchases are simulated; live payments and dining integrations remain future work.

IV. Pricing Structure

A. How Prices Change

Our prototype uses the same pricing process across all three displays. It considers the share of each batch still available and the time left before its selling deadline. More surplus and less time push the target price down, with a default maximum discount of 45%. Food reserved or set aside is excluded from available stock.

Cafe Desk offers 5-, 30-, or 60-minute reviews, with 30 as the default. The usual step toward the target is up to 5% of regular price, rounded up to cents.

For example, a $10 item with half its starting stock still available halfway through the selling window has an $8.88 target. If its current price is $10, the next review shows $9.50, a displayed discount of 5%, because the step is 50 cents, provided no higher minimum applies.

B. Owner Controls and Limits

Cafe owners choose minimum prices and review settings in Cafe Desk. Prices must cover the configured selling costs, fees, and required contribution. A higher minimum can override the usual step limit. Prices can rise as stock falls, even near closing, but never above the regular price. Offers pause if inventory counts are more than 30 minutes old, and new orders stop five minutes before cutoff.

V. Target Users and Customers

Students are the primary users; campus dining operators and cafe managers are prospective customers. After demonstrating operational value, we would test a software subscription priced per location. Expansion would begin at other campus cafes, then other universities with compatible pricing and ordering systems.

A pilot would compare MarketBite with regular prices and a fixed discount under similar conditions. We would track food sold, weighed edible food waste, student savings, cafe revenue, and staff time, accounting for preparation volumes and existing donation commitments. The benefit must exceed discounts and operating costs; the simulation does not establish profitability or waste reduction.

VI. Team Members

Mateo Santalucia - Product and Creative Lead

Jake Gates - Business and Pitch Lead

Joey Kaji - Technical Lead

Markian Dwulit - Sustainability Analyst

Sonia Irakoze - Sustainability Analyst

VII. Code and Supporting Assets

MarketBite Prototype

GitHub Codebase